

Demonstration of STDP Learning Rule using RRAM

EE658 Memory Devices and Technology

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Why Mimic the Brain in Hardware?

The brain performs $\sim 10^{16}$ synaptic operations per second at roughly 20W — a feat no conventional Von Neumann architecture can match. Neuromorphic engineering bridges neuroscience and hardware design, embedding learning directly into silicon.

Parallelism

Billions of neurons fire simultaneously, enabling massive real-time computation.

Co-located Memory

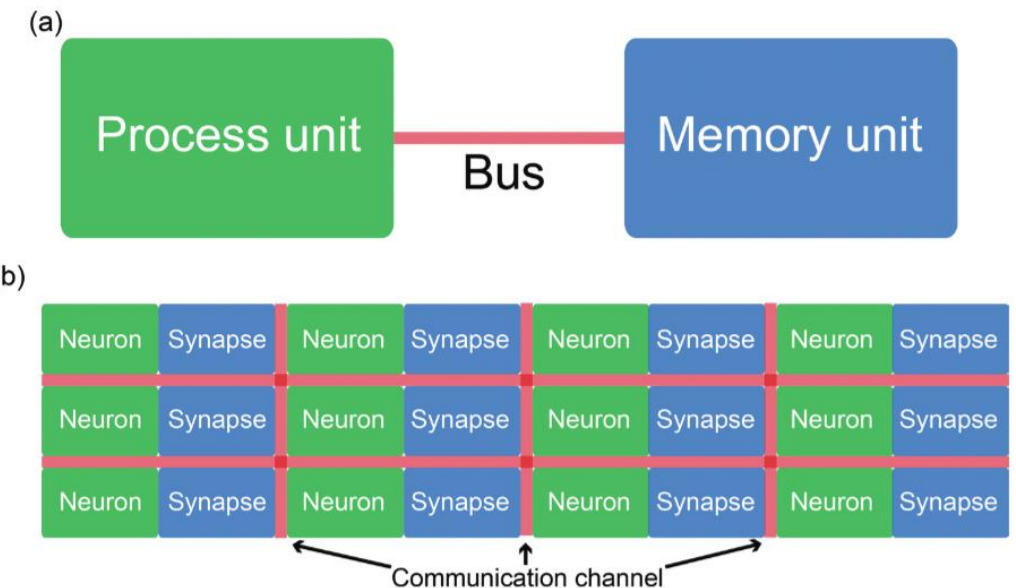
Memory and computation share the same physical substrate — no data movement bottleneck.

The Synapse

The core unit of learning: a tunable connection whose strength encodes information.



Neuromorphic Computing inspired by the Human Brain, synapses and neurons [1].



a) Von Neumann Architecture b) Neuromorphic Architecture [2]

Biological Foundation: Spike-Timing-Dependent Plasticity

Temporal Connection Strengthens Synapses

Hebb's rule: synapses strengthen when pre- and post-synaptic neurons fire in close temporal proximity. STDP formalises this through the timing difference $\Delta = t_{post} - t_{pre}$.

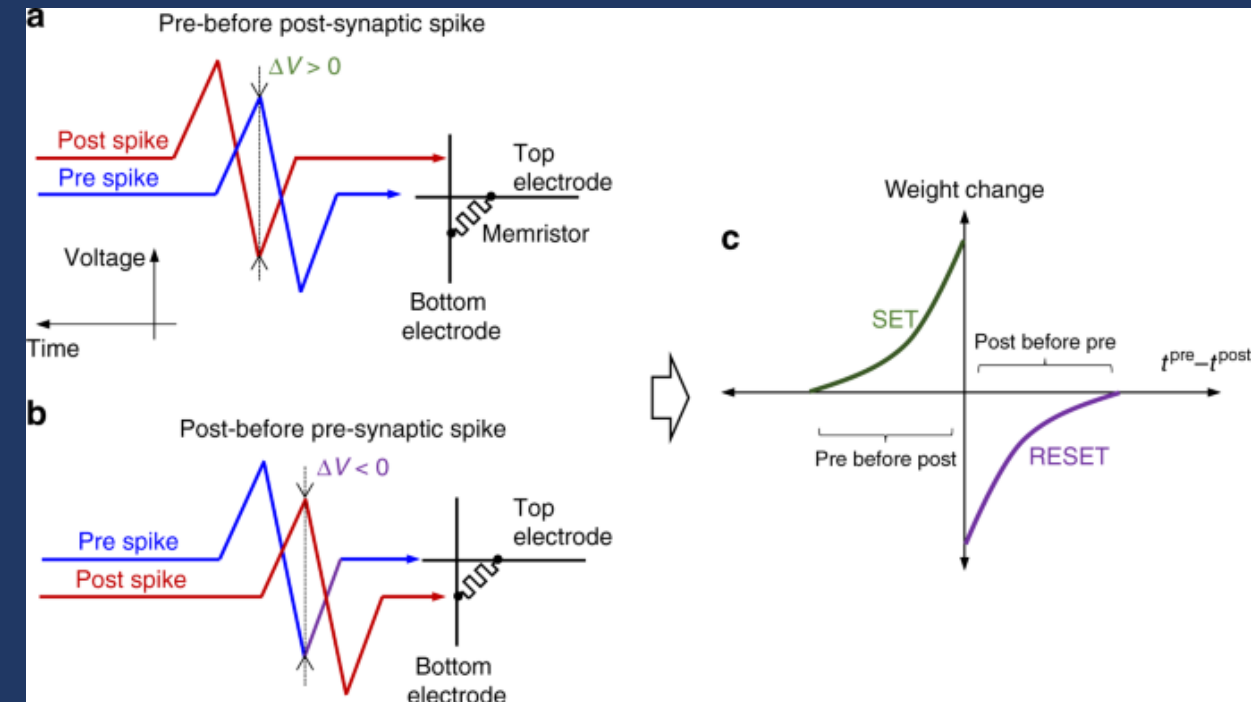
This learning rule can be implemented in hardware using resistive networks: RRAM

LTP — Long-Term Potentiation

Pre fires **before** post ($\Delta t > 0$): synapse strengthens.

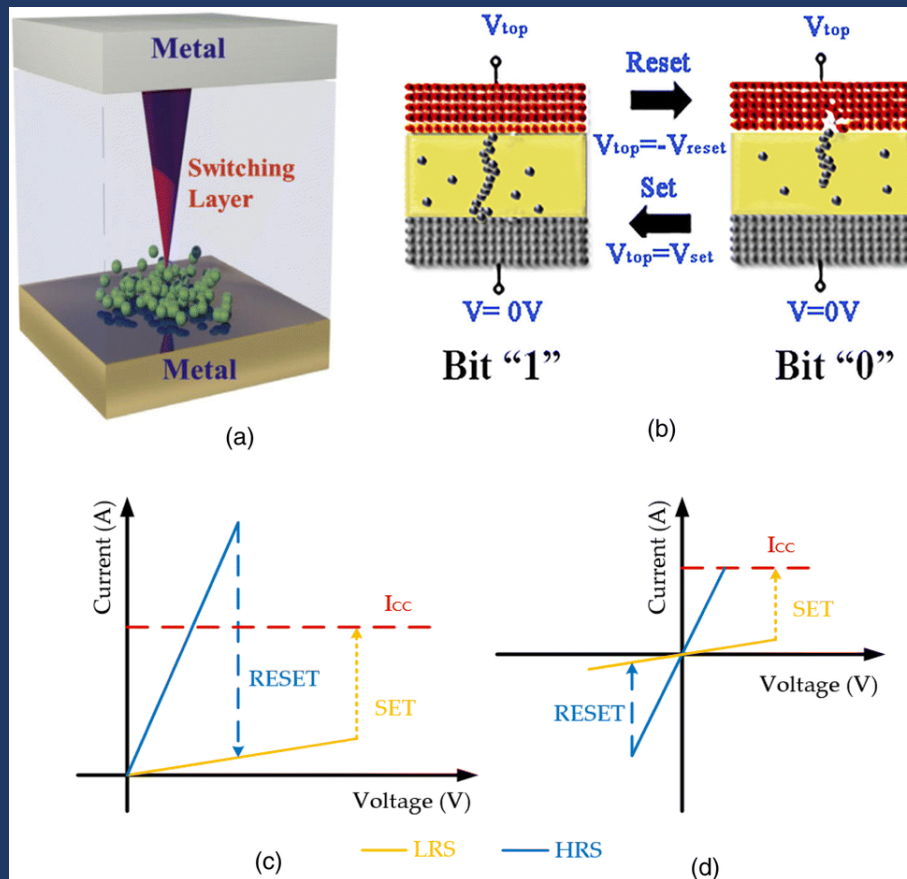
LTD — Long-Term Depression

Pre fires **after** post ($\Delta t < 0$): synapse weakens.



STDP implementation. Relative timing of the pre- and post-synaptic spikes for performing a potentiation and **b** depression of the synaptic weight for the memristor-based implementation¹¹, resulting in **c** the specific STDP, which is commonly observed in biology [3].

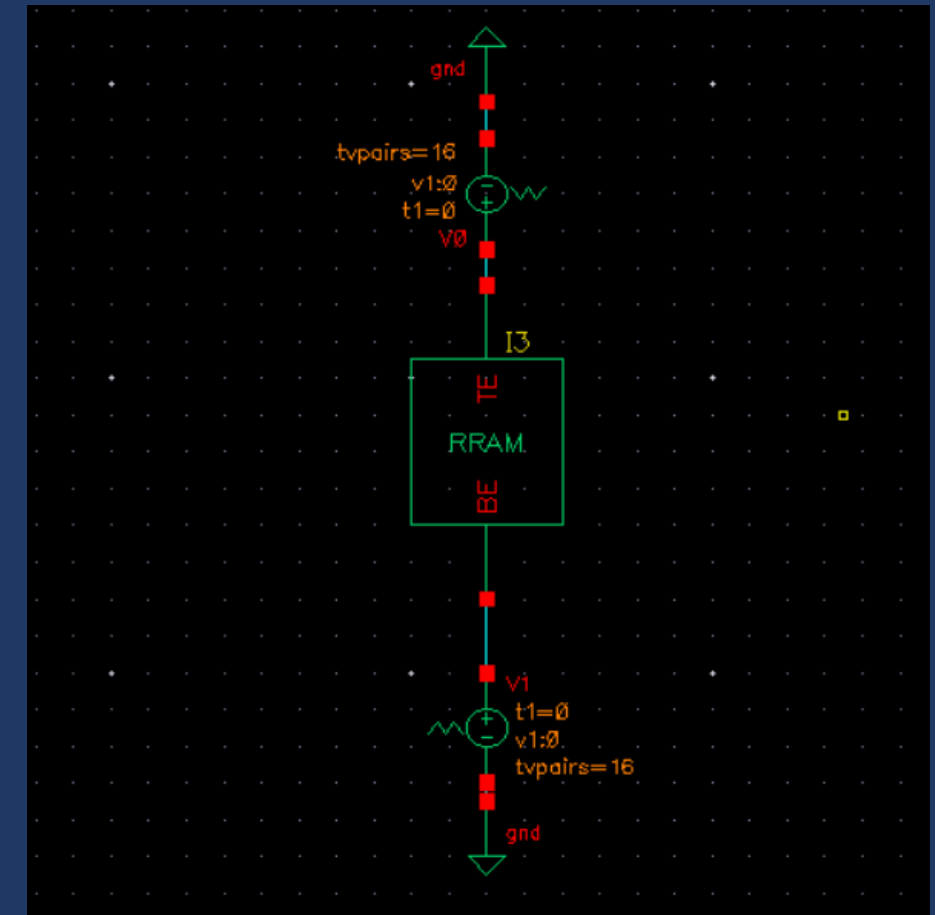
Hardware Implementation: The RRAM Cell



a) Cross-sectional view of an RRAM.
b) Set/Reset Process c) Unipolar switching
d) bipolar switching I - V curve in an RRAM [4].

Resistive RAM as a Memristor

RRAM is a two-terminal device whose resistance depends on the history of applied voltage — a physical realization of the memristor. Conductance G is modulated by the growth and rupture of a conductive filament within a metal-oxide layer.



Designed custom RRAM Schematic

→ SET process: Positive voltage grows the filament → low resistance state (LRS)

→ RESET process: Negative voltage ruptures the filament → high resistance state (HRS)

→ Analog switching: Partial filament growth enables continuous conductance levels

Input Stimulus Design: Bio-Inspired VPWL

Spikes are modeled using VPWL (Voltage Piecewise Linear) sources that replicate the asymmetric shape of biological action potentials, enabling precise Δt overlap for STDP.

Active Head — 1.2V

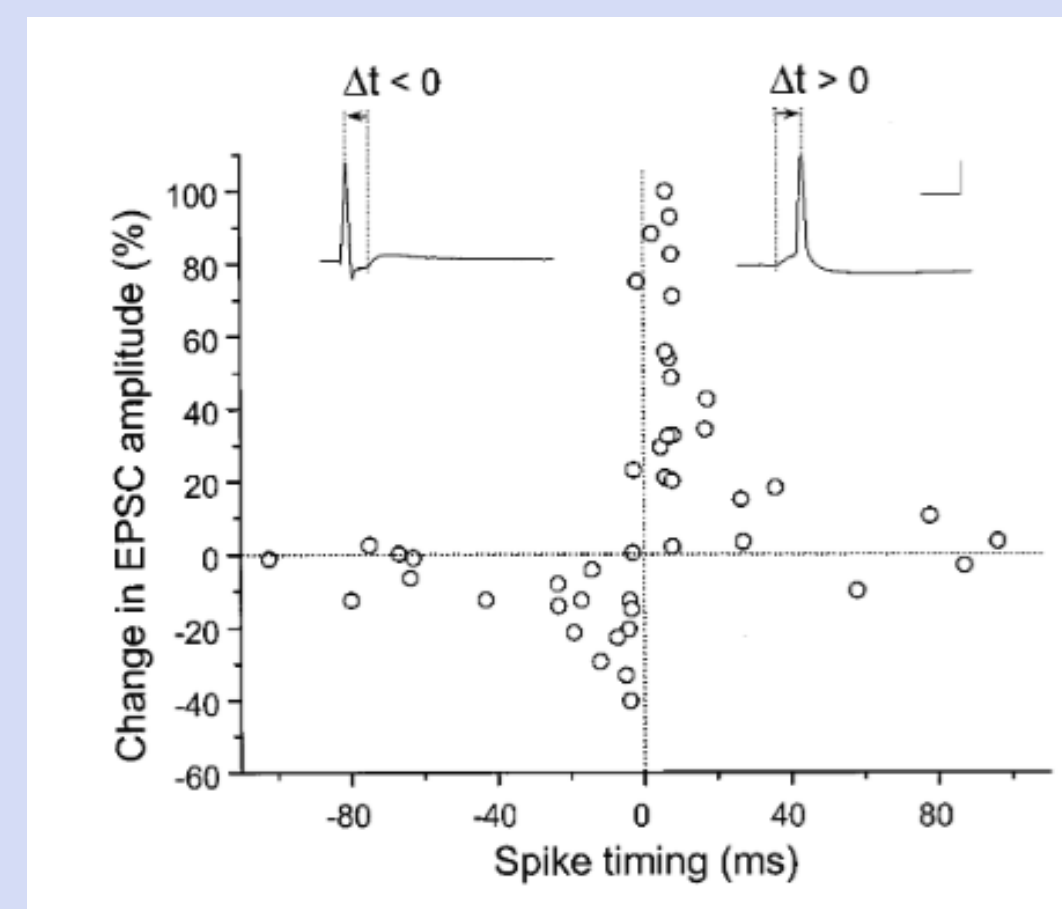
Sharp rising edge exceeds the RRAM threshold voltage V_{th} , triggering filament modulation. Acts as the "spike" event.

Decaying Tail

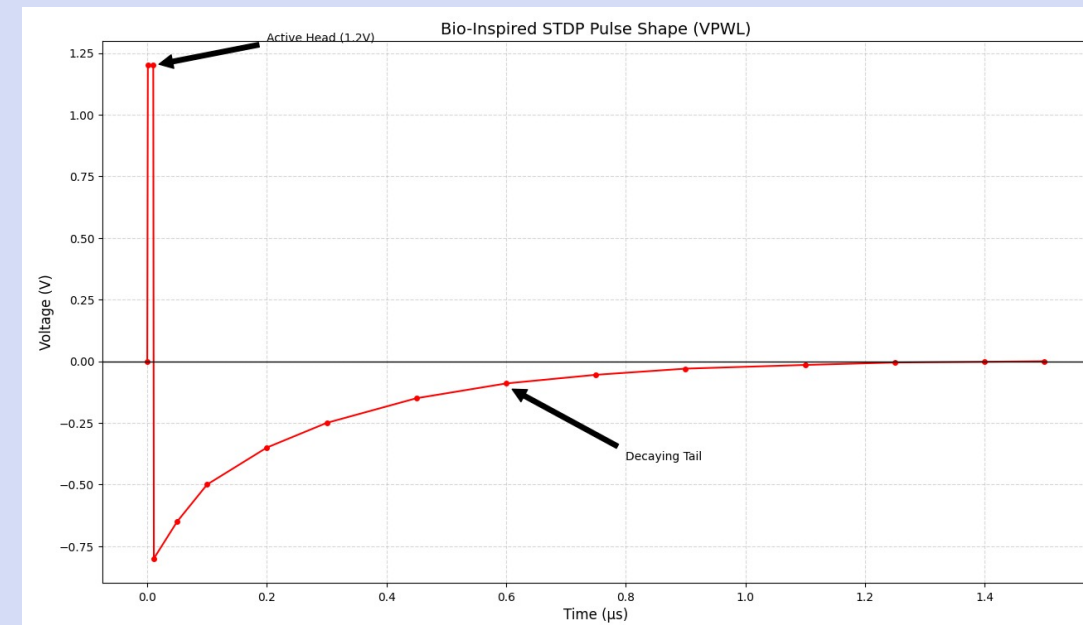
Gradual exponential fall creates a timing-sensitive overlap window. The degree of overlap between pre/post tails encodes $|\Delta t|$.

Timing Overlap

The integral of overlapping tails determines net conductance change — directly mapping biological STDP curves onto hardware.
Mathematical Intuition: $\Delta G \propto \exp(-|\Delta t| / \tau)$



Input Piecewise Linear Voltage function [5]

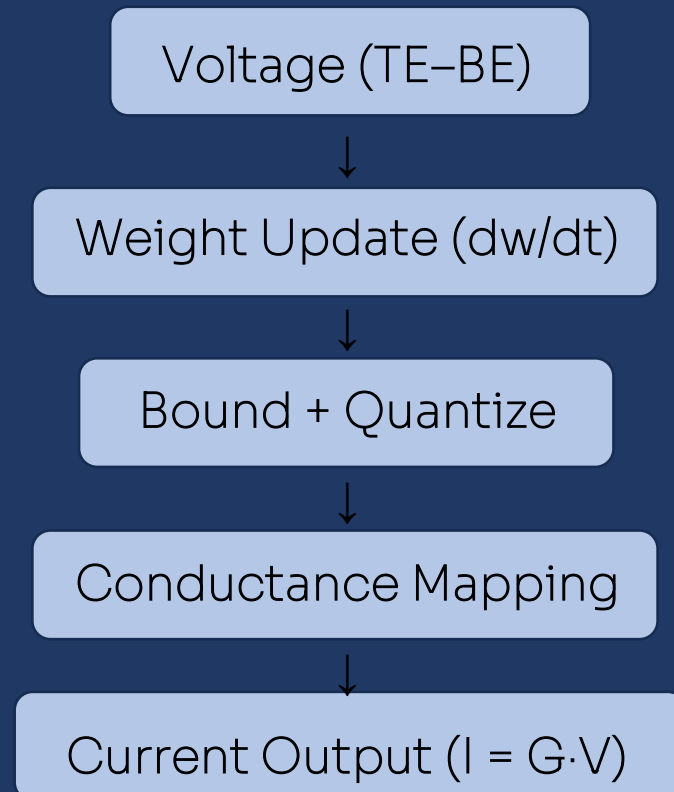


Generated Piecewise Linear Voltage function

Verilog-A Modeling Logic

State Variable Pipeline

Three cascaded state variables implement the full STDP learning rule within a single Verilog-A module.



$$w(t) = \int (dw/dt) dt$$
$$G = w \cdot (1/RL) + (1 - w) \cdot (1/RH)$$
$$I = G \cdot V$$

- Multi-level conductance (256 states)
- Models LRS and HRS switching
- Incremental weight updates
- Enables synaptic learning behaviour.

```
//VerilogA for Memory_Project, RRAM_Cell, veriloga
`include "constants.vams"
`include "disciplines.vams"

module RRAM_Cell(TE, BE);

inout TE, BE;
electrical TE, BE;

//Physical Parameter
parameter real rl = 40;           // Low Resistance State (LRS)
parameter real rh = 13000;        // High Resistance State (HRS)
parameter real vh = 1.5;          // Set Threshold (Positive)
parameter real vl = -1.5;         // Reset Threshold (Negative)

//Learning Parameters
// Note: High alpha is needed to overcome quantization "dead zones"
parameter real alpha_p = 1e7;
parameter real alpha_n = 1e7;
parameter real init_w = 0.2;      // Starting weight (0.0 to 1.0)

//Digital Parameter
parameter integer levels = 1024; // Number of discrete states (e.g., 16 = 4-bit)

// Internal State Variables
real raw_w;
real bounded_w;
real quantized_w;
real G;
real dw_dt;

analog begin

    // Initialize the weight at the start of simulation
    @(initial_step) begin
        raw_w = init_w;
        dw_dt = 0.0;
    end

    // 1. DYNAMIC UPDATE (The "Analog" movement)
    // Determines speed of change based on voltage over threshold
    if (V(TE, BE) > vh && raw_w < 1.0) begin
        dw_dt = alpha_p * (V(TE, BE) - vh);
    end
    else if (V(TE, BE) < vl && raw_w > 0.0) begin
        dw_dt = -alpha_n * (vl - V(TE, BE));
    end
    else begin
        dw_dt = 0.0;
    end

    // 2. ACCUMULATION (Integration)
    // Uses the simulator's time-steps to track the current filament position
    raw_w = idt(dw_dt, init_w);

    // 3. HARD BOUNDS
    // Prevents the filament from growing past physical limits
    if (raw_w > 1.0) bounded_w = 1.0;
    else if (raw_w < 0.0) bounded_w = 0.0;
    else bounded_w = raw_w;

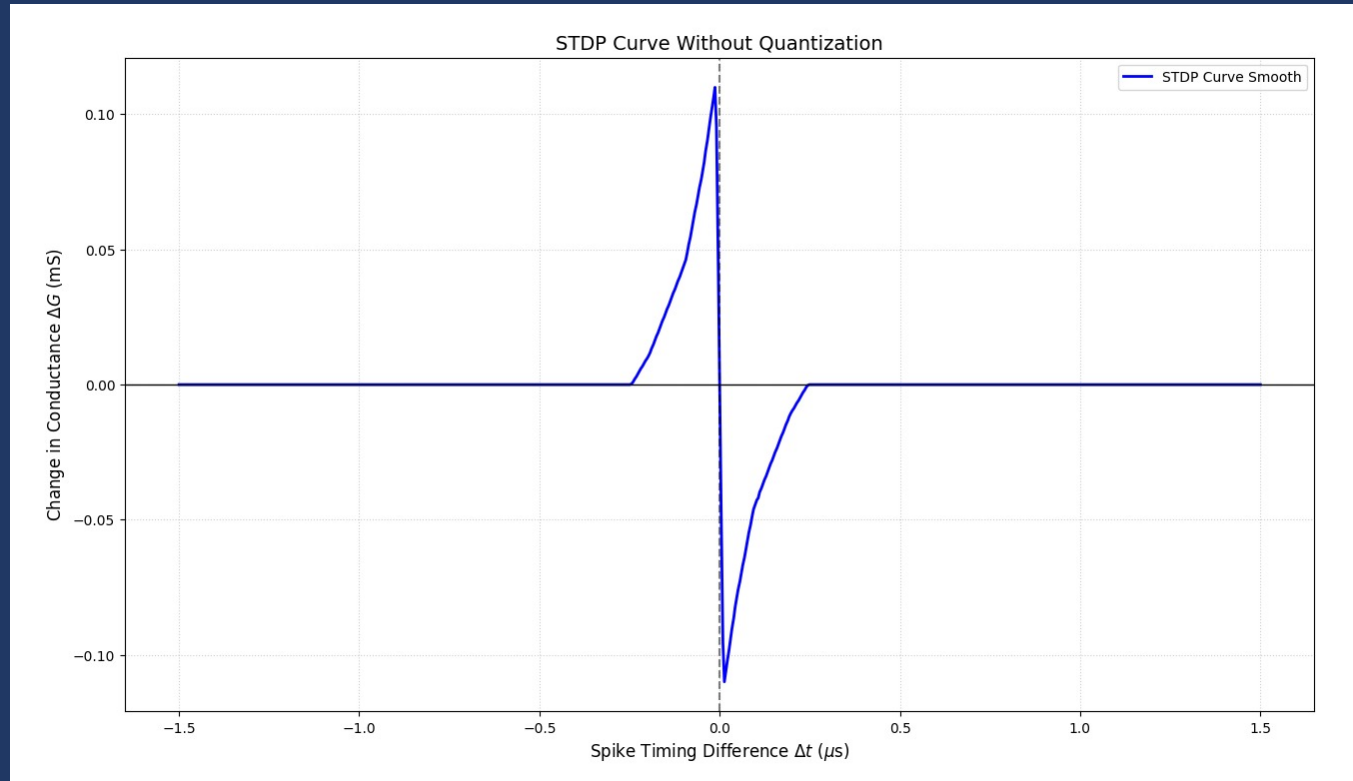
    // 4. THE DIGITAL QUANTIZATION
    // Forces the continuous weight into discrete "buckets"
    // Formula: floor(w * (N-1) + 0.5) / (N-1)
    quantized_w = floor(bounded_w * (levels - 1) + 0.5) / (levels - 1);

    // 5. CONDUCTANCE MAPPING
    // Calculates the final conductance based on the "Digital" weight
    G = quantized_w * (1.0/rl) + (1.0 - quantized_w) * (1.0/rh);

    // Output the current based on the quantized conductance
    I(TE, BE) <+ G * V(TE, BE);

end
endmodule
```

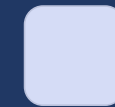
Results: Ideal ΔG vs Δt Curve



The following graph was generated using 701 values of Δt between 0.5 to 3.5 μs .

$$\Delta t = t_{pre} - t_{post}$$

Key Observations



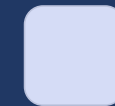
Exponential decay away from $\Delta t = 0$:

As $|\Delta t|$ increases, ΔG approaches zero, meaning distant spikes have negligible influence on learning.



Symmetric Learning Window

LTP and LTD lobes are well-balanced, confirming correct $|\Delta t|$ centering and VPWL overlap logic.



Initial Filament Condition

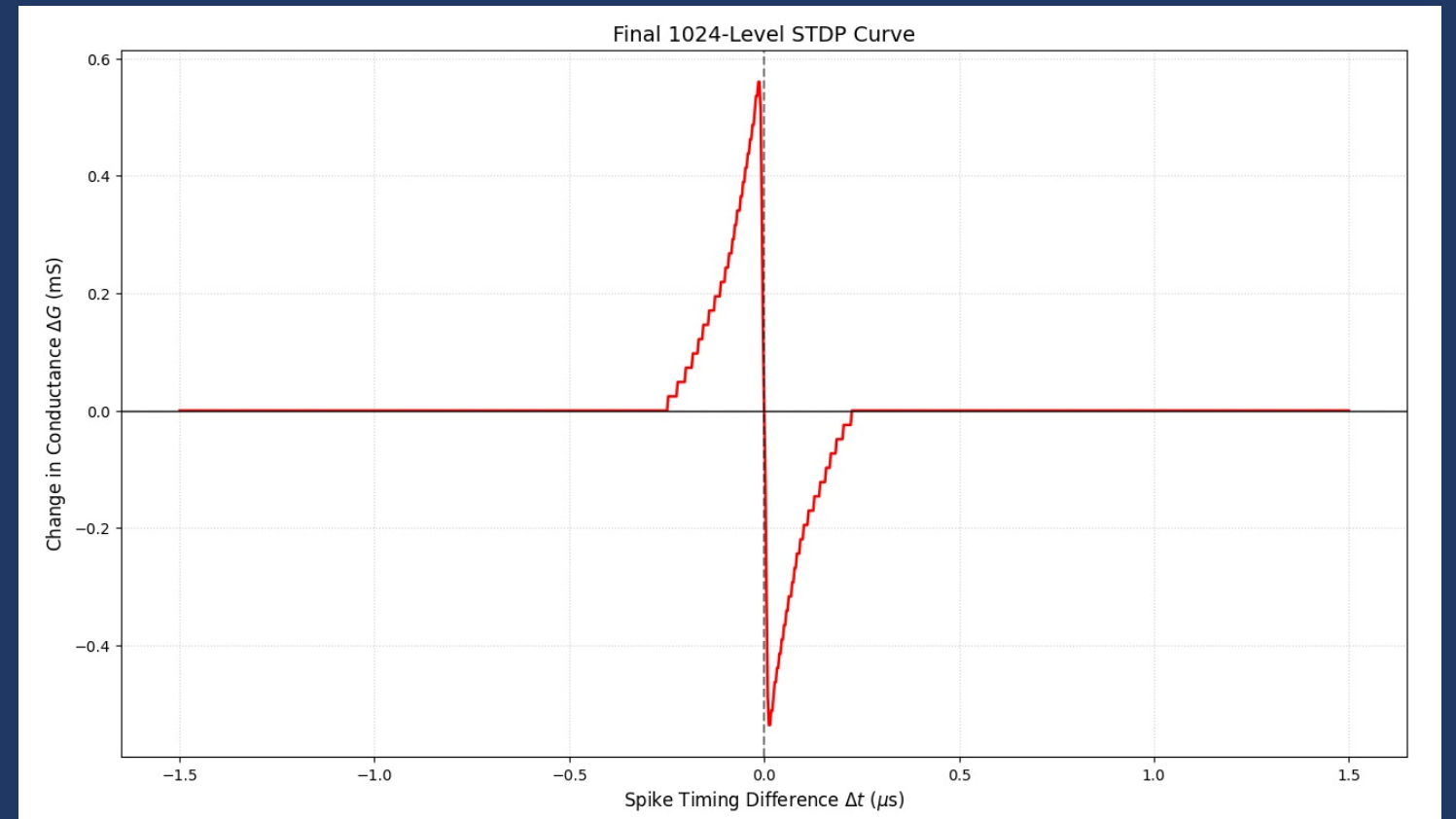
Simulation has been started at 50% filament to create symmetrical graph.

Results: Quantized Graph

From Analog Conductance to Digital Levels

Physical RRAM devices exhibit **continuous conductance**, but digital systems require discrete weight representations. This presentation bridges the gap through a **10-bit quantisation scheme** that maps analogue G to 1024 discrete levels.

High density of states helps minimize **quantization noise**, which can otherwise lead to instability during the neural network training.



10-Bit Architecture

$2^{10} = 1024$ levels spanning G_{min} to G_{max}

Quantization Step

$\Delta G = (G_{min} \text{ to } G_{max}) / 1023$

Rounding Logic

Nearest-level quantization preserves STDP curve fidelity

Conclusion & Future Scope

This work demonstrates a **Verilog-A framework** for STDP learning in RRAM-based synapses, bridging biological accuracy with digital storage constraints.

Biological Fidelity

VPWL-driven STDP faithfully reproduces the LTP/LTD learning window in hardware.

Digital Integration

10-bit quantization enables deployment in digital neuromorphic systems with minimal precision loss.

Edge AI Applications

Scalable to crossbar arrays for on-chip learning in low-power edge devices.

Future Directions

Multi-level cell variability modeling, array-level simulations, and FPGA co-emulation.

References

- [1]“Neuromorphic Hardware and Computing 2024,” Nature, May 06, 2024. Available: <https://www.nature.com/collections/jaidjgeceb/content/brain-inspired-computing-and-algorithms>. [Accessed: Apr. 19, 2026]
- [2] S. Choi, S. Ham, and G. Wang, “Memristor Synapses for Neuromorphic Computing,” *IntechOpen eBooks*, Mar. 2019, doi: <https://doi.org/10.5772/intechopen.85301>
- [3]M. Prezioso *et al.*, “Spike-timing-dependent plasticity learning of coincidence detection with passively integrated memristive circuits,” *Nature Communications*, vol. 9, no. 1, Dec. 2018, doi: <https://doi.org/10.1038/s41467-018-07757-y>
- [4]Furqan Zahoor *et al.*, “An overview of critical applications of resistive random access memory,” *Nanoscale Advances*, Jan. 2024, doi: <https://doi.org/10.1039/d4na00158c>. Available: <https://pubs.rsc.org/en/content/articlelanding/2024/na/d4na00158c>. [Accessed: Nov. 08, 2024]
- [5] G. Bi and M. Poo, “Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type,” *The Journal of Neuroscience*, vol. 18, no. 24, pp. 10464–10472, Dec. 1998, doi: <https://doi.org/10.1523/jneurosci.18-24-10464.1998>. Available: <https://www.jneurosci.org/content/jneuro/18/24/10464.full.pdf>